

# Adaptive Credit-Weighted Online Mirror Descent for Sparse Verifiable Reward Reinforcement Learning

RC-OMD Project Team

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## Abstract

Sparse terminal rewards make it difficult to decide which actions in a long trajectory deserve a large policy update. Broadcasting one group-relative advantage to every decision is inexpensive, but finite-sample correlations can move policy mass at decisions that do not affect the reward. We study a controlled sequence setting in which the reward-relevant positions are known for evaluation but hidden from the optimizer. We introduce Online RC-OMD, a low-cost reliability-calibrated variant of Online Mirror Descent (OMD). It tracks exponentially weighted first and second moments of group-relative action-score estimates and uses their persistent signal-to-uncertainty ratio to scale the local KL-mirror step. The method requires one score computation per group and  $O(H|\mathcal{A}|)$  persistent state. In a frozen out-of-distribution protocol covering four threshold-reward tasks and ten seeds per task, Online RC-OMD remained within 0.0018 normalized success AUC of a tuned uniform Group OMD baseline while using only 18.9%–24.8% as much absolute cumulative KL at known distractor positions. A subsequent frozen shared-linear-policy test passed only the separable-feature scenario and returned No-Go under partial or complete feature aliasing. The combined results support a narrow claim about controlled policy movement and identify parameter coupling as the next bottleneck; they do not establish causal credit assignment or neural-policy performance.

## 1 Introduction

Reinforcement learning with verifiable rewards (RLVR) supplies an attractive training signal for tasks whose final outcomes can be checked automatically. The reward is often binary and observed only after a complete trajectory. A group-relative baseline can avoid learning a separate critic, as in Group Relative Policy Optimization (GRPO) [3], but the same trajectory-level advantage is then commonly propagated through many decisions. This leaves an optimizer-level question: when the available credit signal is a noisy proxy, how much policy movement should each decision be allowed to make?

Recent work has attacked token-level credit assignment using entropy-aware weights and related local signals [2, 5]. Entropy can be informative, but uncertainty is not identical to reward relevance. A decision can be uncertain and irrelevant, or confident and pivotal. We therefore treat credit reliability as a quantity to be estimated rather than equating it with entropy. Our approach is deliberately smaller than language-model RL: a factorized tabular policy lets us observe local policy movement exactly and construct environments with known distractor decisions.

Our contributions are:

1. We construct controlled sparse terminal-reward sequence tasks with exact counterfactual credit and known reward-irrelevant positions for evaluation. The suite includes settings where policy entropy is aligned with, and deliberately misleading about, true positional importance.
2. We propose Online RC-OMD, which converts persistent agreement in online group-relative action scores into per-position OMD step scales. Unlike a per-group bootstrap estimator, it adds no rollout resampling and maintains only  $O(H|\mathcal{A}|)$  state.
3. We report development results separately from a protocol frozen before evaluation. The pre-registered comparison passes all four unseen scenarios under a declared AUC–distractor-KL rule.
4. We run a second frozen transfer test with shared linear policy parameters. It preserves the result only when pivotal and distractor features are separable, exposing an explicit limit of local reliability scaling followed by global parametric projection.

The central empirical claim is a trade-off, not universal dominance: reliability calibration directs substantially less policy movement toward known distractors while retaining almost all learning efficiency in the tested controlled tasks.

## 2 Related Work

**Mirror-descent policy optimization.** Mirror descent replaces Euclidean proximity with a Bregman divergence and is particularly natural on a probability simplex [1]. MDPO connects this geometry to practical trust-region policy optimization by combining a linearized RL objective with a proximity term [4]. We use the negative-entropy mirror map, for which the proximity term is KL and the exact local update is exponentiated gradient.

**Group-relative optimization and sparse verifiable rewards.** DeepSeekMath introduced GRPO as a critic-free relative policy optimization method for mathematical reasoning [3]. Outcome-only feedback saves the cost of a process reward model, but it leaves temporal credit coarse. A recent survey organizes the rapidly growing LLM credit-assignment literature by granularity and methodology [6].

**Entropy-aware credit.** EAPO analyzes reward polarity and token entropy, and shows that entropy bounds the credit a token can carry under its information-theoretic formulation [2]. ACPO uses a mode-local surrogate entropy to modulate token updates asymmetrically [5]. Our study does not compete with these systems at language-model scale. Instead, it tests a distinct diagnostic hypothesis: persistent score agreement can regulate local mirror geometry even when instantaneous entropy is a misleading proxy for reward relevance.

## 3 Controlled Problem

Let a trajectory be  $\tau = (a_1, \dots, a_H)$  with  $a_k \in \mathcal{A} = \{1, \dots, A\}$ . The factorized policy is  $\pi(\tau) = \prod_{k=1}^H \pi_k(a_k)$ . A hidden set  $\mathcal{K} \subseteq \{1, \dots, H\}$  contains the pivotal positions. Each  $k \in \mathcal{K}$  has a target action  $a_k^*$ . The terminal reward is

$$R(\tau) = \mathbf{1} \left\{ \sum_{k \in \mathcal{K}} \mathbf{1}\{a_k = a_k^*\} \geq m \right\}. \quad (1)$$

Positions outside  $\mathcal{K}$  are distractors: changing them never changes the reward. The special case  $m = |\mathcal{K}|$  is an all-match or “needle” task. Threshold tasks use  $m < |\mathcal{K}|$ .

For each update,  $K$  trajectories are sampled and assigned group-relative advantages

$$\hat{A}_i = R(\tau_i) - \frac{1}{K} \sum_{j=1}^K R(\tau_j). \quad (2)$$

We estimate the gain for action  $a$  at position  $k$  with inverse propensity weighting,

$$\hat{g}_{t,k,a} = \frac{1}{K} \sum_{i=1}^K \frac{\mathbf{1}\{a_{i,k} = a\} \hat{A}_i}{\max(\pi_{t,k}(a), p_{\min})}, \quad (3)$$

with a fixed importance cap in the implementation. Uniform Group OMD applies this noisy action score at every position. Although the expected distractor score is zero in our controlled environment, its finite-group realization need not be, so repeated updates may accumulate irrelevant KL drift.

For diagnosis only, the known transition and reward structure gives exact on-trajectory counterfactual credit

$$c_k(\tau, \pi) = Q^\pi(s_k, a_k) - V^\pi(s_k). \quad (4)$$

The optimizer never receives the pivotal set or oracle credit unless explicitly labelled as an oracle baseline.

## 4 Reliability-Calibrated OMD

### 4.1 Uniform KL-mirror update

At position  $k$ , negative-entropy OMD solves

$$\pi_{t+1,k} = \arg \max_{p \in \Delta_A} \{ \eta \langle \hat{g}_{t,k}, p \rangle - D_{\text{KL}}(p \| \pi_{t,k}) \}, \quad (5)$$

with solution

$$\pi_{t+1,k}(a) = \frac{\pi_{t,k}(a) \exp(\eta \hat{g}_{t,k,a})}{\sum_b \pi_{t,k}(b) \exp(\eta \hat{g}_{t,k,b})}. \quad (6)$$

### 4.2 Online reliability estimate

We center each score vector across actions and maintain exponentially weighted first and second moments with decay  $\beta$ . Let  $\hat{\mu}_{t,k}$  be the running mean vector,  $\hat{s}_{t,k}$  its estimated standard-error vector, and  $n_{\text{eff},t}$  the effective sample size of the exponential weights. We define

$$q_{t,k} = \min \left\{ 1, \frac{n_{\text{eff},t}}{n_{\text{warm}}} \right\} \left[ 1 - z \frac{\|\hat{s}_{t,k}\|_2}{\max(\|\hat{\mu}_{t,k}\|_2, \epsilon)} \right]_+, \quad (7)$$

$$\alpha_{t,k} = \alpha_{\min} + (1 - \alpha_{\min})q_{t,k}. \quad (8)$$

The first factor prevents confidence before enough effective observations; the second rewards a persistent signal relative to its uncertainty. The local OMD update replaces  $\eta$  in Eq. (6) by  $\eta\alpha_{t,k}$ . The current score determines direction, whereas the historical reliability estimate controls how far the policy may move.

**Proposition 1** (Exact local movement control). *For any actions  $a, b$  with positive probability, the reliability-calibrated update satisfies*

$$\log \frac{\pi_{t+1,k}(a)}{\pi_{t+1,k}(b)} - \log \frac{\pi_{t,k}(a)}{\pi_{t,k}(b)} = \eta\alpha_{t,k}(\hat{g}_{t,k,a} - \hat{g}_{t,k,b}). \quad (9)$$

Thus reliability scales every local log-odds displacement linearly, without changing the ranking induced by the current action-score vector.

*Proof.* Take the log ratio of the two normalized exponentiated-gradient probabilities; the shared normalizer cancels.  $\square$

The estimator uses one action-score computation per group. Its first and second moments require  $O(HA)$  memory and  $O(HA)$  arithmetic per update, matching the asymptotic score-update cost of the uniform baseline.

## 5 Experimental Design

We separate exploratory development from frozen evaluation. Development tasks were used to construct entropy counterexamples, compare bootstrap and online reliability, and choose the online configuration. The subsequent OOD protocol, including tasks, seeds, metrics, parameter values, and decision rule, was committed before execution under identifier `ood-v1-2026-08-08`.

The frozen comparison used ten seeds per task. Uniform Group OMD used base step 0.75. Online RC-OMD used base step 1.25, decay 0.9,  $z = 1$ , warm-up 8, and reliability floor 0.1. These unequal base steps were selected on development tasks to compare a Pareto-matched operating point; a separate same-step diagnostic was also frozen. Normalized success AUC is the mean exact success probability across evaluation checkpoints. Distractor KL is the sum over training iterations and known distractor positions of the absolute local KL movement.

A scenario passed when Online RC-OMD was no more than 0.01 below Uniform in success AUC and used at most 50% of Uniform’s distractor KL. The method-level Go decision required at least three of four scenario passes. Runtime feasibility required an Online-to-Uniform ratio below 1.5 in at least three scenarios.

The second frozen protocol replaces independent policy rows with shared linear logits  $\ell_k = \phi_k^\top W$ . Each method first forms a tabular OMD target and then performs 60 fixed gradient steps minimizing average target-to-parametric KL. Three scenarios use separable, partially aliased, and completely aliased position features. Projected transfer required at least two of three scenarios to remain within 0.02 AUC and 75% distractor KL of Projected Uniform.

Table 1: Frozen OOD comparison. AUC difference and KL ratio are Online minus Uniform and Online divided by Uniform, respectively. Values are means across ten seeds; the seed is the replication unit.

| Scenario         | Uniform AUC | Online AUC | $\Delta$ AUC | KL ratio | Runtime ratio |
|------------------|-------------|------------|--------------|----------|---------------|
| Dense 2-of-6     | 0.992182    | 0.990865   | -0.001317    | 0.248    | 1.107         |
| Needle 5-of-5    | 0.980026    | 0.980782   | 0.000755     | 0.239    | 1.081         |
| Threshold 3-of-5 | 0.992195    | 0.990567   | -0.001628    | 0.189    | 1.069         |
| Threshold 4-of-6 | 0.989446    | 0.987688   | -0.001758    | 0.199    | 1.079         |

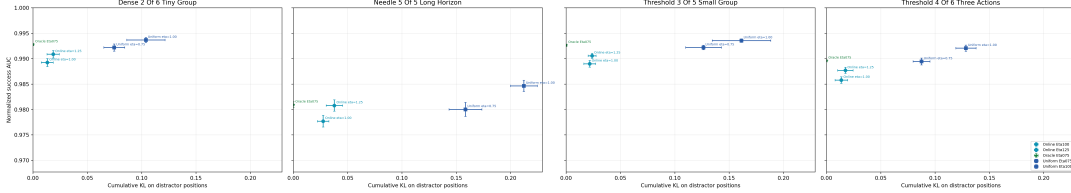


Figure 1: Pre-registered OOD success–distractor-drift comparison. The generated artifact is reproduced from the frozen configuration; the numerical decision is based on the machine-readable summary rather than the figure.

## 6 Results

### 6.1 Pre-registered OOD evaluation

All four scenarios passed the pre-declared rule (Table 1), yielding a Go decision. Online RC-OMD retained AUC to within 0.0018 while using between 18.9% and 24.8% as much distractor KL. Runtime overhead was 6.9%–10.7%, so all four scenarios also passed the systems rule. The scenarios changed reward structure, horizon, group size, initial success probability, and included both two- and three-action policies.

At a common base step of 1.0, Online RC-OMD had 0.0044–0.0069 lower AUC, but used only 10.7%–13.4% as much distractor KL. This diagnostic confirms that the main result is an update-allocation trade-off, not evidence that reliability scaling always raises reward at an identical nominal step size.

### 6.2 Development diagnostics

In entropy-aligned development tasks, entropy weighting improved early learning; when distractor positions were initialized with high entropy and pivotal positions with lower entropy, the same heuristic reduced sample efficiency. This controlled reversal motivated reliability estimation. Per-group bootstrap reliability reduced distractor drift but cost roughly 7–9 $\times$  the uniform runtime. Replacing bootstrap resampling with online moments reduced runtime to within roughly 7%–13% of Uniform on the development tasks. These measurements are exploratory and are not pooled with the frozen OOD result.

### 6.3 Frozen shared-parameter transfer

All three AUC conditions pass, but only the separable-feature task passes the KL condition (Table 2). The estimator itself continues to rank pivotal positions above distractors: mean pivotal reliability is 0.401–0.432, mean distractor reliability is 0.137–0.167, and top- $k$  precision is 0.704–0.725. The failure occurs after projection. With complete aliasing, every pivotal position shares its feature with a distractor, so their projected KL movements are structurally identical. The larger Online base step then produces 17.6% more distractor KL than Projected Uniform. The runtime ratio remains 0.999–1.055 across the three scenarios.

## 7 Discussion and Limitations

The results are consistent with a simple mechanism: reward-relevant score directions recur, whereas finite-sample distractor directions fluctuate, so running agreement can allocate a larger KL budget to the former. This is an associational mechanism, not a causal-credit theorem. A persistent spurious correlation can also look reliable, and estimator lag can suppress newly useful directions after a policy or task change.

Table 2: Frozen shared-linear-policy transfer. Only the separable scenario passes the declared AUC and distractor-KL conditions; the resulting transfer decision is No-Go.

| Feature geometry  | Uniform AUC | Online AUC | $\Delta$ AUC | KL ratio | Pass |
|-------------------|-------------|------------|--------------|----------|------|
| Separable         | 0.982550    | 0.980943   | -0.001608    | 0.193    | Yes  |
| Partial aliasing  | 0.977562    | 0.977033   | -0.000529    | 0.967    | No   |
| Complete aliasing | 0.971513    | 0.972722   | 0.001209     | 1.176    | No   |

The shared-linear transfer result sharpens the mechanism: the reliability estimate can remain discriminative while the parameterization prevents its local constraints from being realized. Reliability must therefore enter the parameter-space metric or constrained projection, rather than only the tabular target distribution.

The current scope has five important limits. First, function approximation is restricted to fixed linear features; no representation is learned. Second, known distractor positions are used only to evaluate drift, an oracle unavailable in real RLVR. Third, the principal methods use base steps chosen on development tasks. Fourth, ten seeds quantify simulation variation but not universality. Fifth, no nonlinear neural policy, language model, process reward model, or long natural-language trajectory has been tested. These limitations rule out a direct language-model experiment with the current optimizer.

## 8 Conclusion

We studied whether the reliability of a noisy step-level credit proxy should control local mirror-descent geometry under sparse terminal rewards. A low-cost online moment estimator produced a reproducible Pareto improvement in four frozen controlled OOD scenarios: nearly unchanged success learning with substantially less policy movement at known distractor decisions. A frozen shared-linear transfer test then showed that this behavior survives separable features but not feature aliasing. The next algorithmic stage is to make credit reliability control parameter-space geometry itself, then subject that new method to a separately frozen transfer protocol.

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